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Development of a body of knowledge for design for disassembly and recycling of high-tech products: a case study on lithium-ion batteries

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ABSTRACT

Demand for electronic products is growing, as is the volume of waste electrical and electronic equipment (WEEE). To reduce their environmental impact, particularly during their end-of-life, it is important to apply eco-design practices such as design for disassembly (DFD) and design for recycling (DFR) from the beginning of their development. However, these strategies are not systematically implemented by manufacturers due to several challenges, such as the complexity of the methods, the uncertainty and variability of the materials and components, and the lack of knowledge on DFD and DFR. This study aims to develop a body of knowledge (BOK) for DFD and DFR of electronic products to fill this gap. A systematic comparison of different BOKs has led to the proposal of a BOK composed of four main parts: Areas of Knowledge, Tools and Techniques, Skills and Abilities, and Terminology. The proposed framework was applied to lithium-ion batteries (LIBs) as an example of electronic products that require high-tech solutions for their efficient and sustainable management. This approach is essential for high-tech products, as they often contain valuable and scarce materials that need to be recovered and reused in a circular economy. The results showed that the BOK was an effective tool in supporting the sustainable development of batteries.

ARTICLE HISTORY

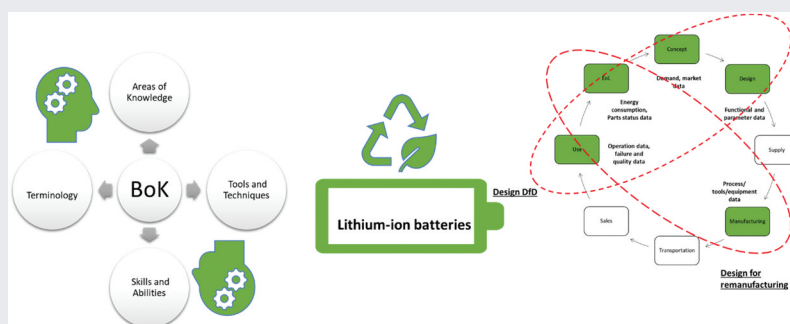
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KEYWORDS

Body of knowledge; design for disassembly; design for recycling; high-tech products; lithium-ion batteries



1. Introduction

The growing demand for electronic products and the rapid development of new technologies have led to a significant increase in electronic waste generation. This growth is a sustainability challenge that needs to be managed properly, especially as these products contain critical and rare materials. Critical materials are elements that are essential for various technological applications, especially in the fields of energy, electronics, and communication. Some examples of rare earth elements are lithium, cobalt, platinum group elements, gallium, germanium, and indium [1]. These materials become an environmental burden when they are not properly recycled or disposed of after the end of their useful life in electronic devices. Moreover, E-wastes can be toxic, are not biodegradable and accumulates in the environment, in the soil,

air, water and living things [2]. They can also release harmful substances such as mercury, lead, cadmium, arsenic, and brominated flame retardants, which can cause irreversible health effects, including cancers, miscarriages and neurological damage [3,4].

One way to address this challenge is to apply eco-design practices from the beginning of product development, to reduce their impact during their entire life cycle. Eco-design is a multidisciplinary field and faces several challenges, such as the lack of knowledge, the difficulty of multidimensional analysis and the trade-offs between different design criteria [5]. DfD and DfR are two eco-design strategies that facilitate EoL management of products and materials. DfD can help reduce the time and cost of disassembly operations by considering different criteria such as the number of mating faces, the type of connections, the accessibility

of components, etc. DfR can help increase the circularity rate and the value of materials by considering the compatibility, purity, and separability of different materials. These strategies can also improve the performance, survivability, scalability, and commonality of products. However, applying DfD and DfR requires a comprehensive and updated knowledge base that covers technical, environmental, and economic aspects of product design and EoL management. Therefore, designers need to use practices and guidelines, to help them acquire and apply relevant knowledge in an effective and efficient way. DfD and DfR are promising approaches to overcome some of the main challenges of eco-design, but they also require a continuous development and integration of knowledge and practices in this field. Future research and practice should focus on creating and improving the tools and methods that can support designers in implementing DfD and DfR in complex products.

To address this gap, this study aims to develop a BOK for DFD and DFR of electronic products, with a particular focus on LIBs.

LIBs are widely used in high-tech products, such as laptops, cell phones, and electric vehicles, due to their high energy density, long cycle life, and low self-discharge rate. However, LIBs have several environmental impacts throughout their life cycle, from extraction and production to use and disposal. According to [6], the extraction process of lithium requires large amounts of water and energy, which can deplete and contaminate local water resources, affect biodiversity, and contribute to greenhouse gas emissions. The production process also involves toxic chemicals and metals which can pose health and safety risks for workers and communities. End of life contains hazardous materials that can leach into soil and water if not properly recycled [7]. The recycling phase also poses several challenges that affect the recovery rate of lithium-ion batteries. Despite these environmental issues, LIBs are widely used as a green technology for reducing carbon emissions. Moreover, the rapid growth of the electric vehicle (EV) market has raised concerns about the sustainable supply of battery materials. These concerns include the limited availability of cobalt and lithium [8], the high geopolitical concentration of cobalt [9,10], the social and environmental impacts of mining for battery materials [11,12], and the need to rapidly scale up battery supply chains to meet demand [10].

Therefore, it is essential to design LIBs for disassembly and recycling, which can reduce the environmental impact, recover valuable materials, and extend the product life cycle. To the best of our knowledge, the current practices of DFD and DFR for LIBs are limited and fragmented, and there is a lack of systematic and comprehensive body of knowledge to guide the designers and recyclers.

A BOK is a comprehensive framework that encompasses the essential information, theories, and practices related to a specific domain or discipline. A BOK can help designers and engineers to acquire the necessary knowledge and skills to apply DFD and DFR principles in their product development process. To develop this BOK, a systematic comparison of various elements extracted from multiple existing BOKs was done. This enabled us to propose an original framework for DFD and DFR, tailored to the unique requirements of electronic products. The proposed framework consists of four main parts: Areas of Knowledge, Tools and Techniques, Skills and Abilities, and Terminology. The framework was then applied to LIBs as a case study, demonstrating its usefulness and effectiveness in supporting the sustainable development of batteries. This application also provides practical examples and guidelines for applying eco-design principles to LIBs.

Based on this methodology, the main contributions of this article are:

- Proposing a BOK framework for DFD and DFR of electronic products
- Applying the proposed framework to create a customized BOK for DFD and DFR of LIBs

The rest of the paper is organized as follows: [Section 2](#) consists of a literature review highlighting the importance of DFD and DFR in the context of high-tech products, by providing the state of art of high-tech products, their end-of-life management, the importance of knowledge, as well as the existing BOKs, to identify gaps and prove the originality of this work. [Section 3](#) describes the methodology for the proposed BOK, outlining its various components and their significance. In [Section 4](#), the practical application of the BOK is demonstrated through a case study on LIBs and the results are discussed. Finally, [Section 5](#) provides the discussions and the implication of the research and finally, [Section 6](#) concludes the article and suggests avenues for future research.

2. Literature review

2.1. High-tech products

High-tech products are those that use advanced technology and have a high level of innovation, complexity, and diversity. They often contain critical and rare materials, have a short lifespan, and generate significant environmental impacts at the end of their life. Examples of high-tech products include electrical and electronic equipment (EEE). The high-tech sector is an important and fast-growing industry in many countries. For instance, in Canada, hi-tech exports increased from 14.7% to 16.1% between 2011 and 2019% of

manufactured exports) [13], and total high-tech products imports and exports in the EU increased from 482 billion euros to 777 billion euros between 2011 and 2021 [14]. Moreover, in 2020, high-tech supplied local economies in the US with over 29% of manufacturing exports [15], and in 2016, the high-tech sector accounted for about 25% of the total U.S. economic output [16].

This paper focuses on EEE that share the following characteristics:

- They are replacing or being integrated into conventional products, batteries or plugs for example, that are being used in an increasing number of products such as smartphones, tablets, and medical equipment [17].
- Hi-tech products usually have a short lifespan, generally due to customers' preferences, and the end-of-life stage is often neglected in the design of these products, resulting in an environmental burden of e-waste [18,19].
- The trend is toward miniaturization of these products (smartphones, tablets, batteries, etc.) and to develop them, designers are using a wide variety of materials, including critical and rare materials. These materials generate significant impacts on the environment and, most of the time, are present in small quantities, making their recovery at the end of their life very difficult [20,21].
- These products present a high level of innovation; the National Center for Science and Engineering Statistics NCSES classifies electronic equipment among the high R&D intensity category [22], the R&D intensity being the ratio of R&D to the added value within an industry [23].
- These products are numerous with a high growth rate, and forecasts indicate that this trend will continue. On average, the total weight of global EEE consumption (excluding photovoltaic panels) is increasing by 2.5 million tons (Mt) per year [24].

2.2. End-of-life and eco-design of hi-tech products

Growing environmental awareness and legislation are prompting manufacturers to consider the products' end-of-life stages [25]. Most of the high-tech products are not designed for a second life cycle, as the fate of about 83% of e-waste generated globally in 2019 was uncertain [24]. The 9 R framework describes 9 strategies for the end-of-life of products, Recover (R9) being the less circular strategy, then recycle (R8), up to Refuse (R0) which is the most circular option [26]. However, we're still in the early stages of circular economy implementation worldwide, mainly focusing on recycling [27]. Furthermore, the efficiency of recycling is dependent on the internal morphology of the product, which is determined in the design phase, and the recycling

measures can be significantly improved by considering disassembly during the design process [28,29]. Also, materials are often recovered based on the profitability of the process and not necessarily for environmental reasons; recycling of LIBs, for example, is mainly economically motivated [30]. In addition, EEE are not assembled nor designed with recycling principles in mind [24], and end-of-life constraints should be incorporated in the early design process [31]. Moreover, besides the development process, competence and motivation are other factors to successfully integrate eco-design in product development. For instance, an environmental specialist could support the development activities, and individuals should be encouraged to actively participate in the integration of eco-design [32]. Also, for a high-tech company to stay competitive, the competencies of its employees should be expanded and improved [33].

2.3. Eco-design and importance of knowledge in DFD and DFR of high-tech products

DFD and DFR are part of Eco-design. In fact, Eco-design includes any type of design that contributes to minimizing the environmental impacts of a product or service throughout its lifecycle, and it is important since 80% of the product's environmental footprint is determined during the design phase [34]. It should also be noted that another important aspect of eco-design is its contribution to the circular economy which is based on the use of renewable resources [35]. In the EEE industry, there are several drivers of eco-design, such as legal requirements and corporate responsibility, and a more circular economy could be achieved with several design strategies such as DFD and appropriate material combinations [36,37].

A study on design for environment and DFD practices among Malaysian manufacturers showed that the decision to integrate environmental aspects into design is mainly based on the knowledge of the designers, and that the lack of appropriate knowledge is a significant cause of the lack of implementation of DFD during the design process [38]. In addition, there is a gap between the treatment of the product at the end of its life and its design phase. To improve circular economy, a circular design approach has also been developed, which considers design for and from recycling [39]. Moreover, there is a lack on the "how" and "who" should implement eco-design in companies, and a knowledge holder is suggested to support the team or organization in the eco-design process [40]. His/her knowledge includes lifecycle analysis LCA and materials and their selection. In addition to LCA, The industry uses technology assessment tools throughout the innovation cycle, such as techno-economic assessments [41].

Therefore, a systematic approach for developing a BOK aids designers to effectively integrate eco-design principles and practices into their work.

2.4. Existing body of knowledge books

A BOK refers to the complete set of information, theories, and practices that are accumulated and organized within a particular field or discipline. To develop this BOK, it is necessary to examine other bodies of knowledge. Several BOKs have been developed in the past, in different fields, and although they are not directly related to eco-design of high-tech products, some common elements can be identified to develop the present BOK, as described in the following paragraphs. The bodies of knowledge identified in the literature include:

- A Guide to the Project Management Body of Knowledge (PMBOK® Guide) [42]
- Information Management Body of Knowledge (IMBOK) [43]
- Guide to Software Engineering Body of Knowledge (SWEBOK) [44]
- Environmental Engineering Body of Knowledge (EnvE BOK) [45]
- Body of Knowledge for Software Quality Measurement [46]
- Industrial and systems engineering (IISE) body of knowledge [47]
- The Personal Software ProcessSM (PSPSM) Body of Knowledge [48]
- Professional Engineering Body of Knowledge [49]

This literature review will explore the PMBOK® Guide, IMBOK, SWEBOK, and EnvE BOK in more detail, as they cover a variety of areas and are well-developed and established in the scientific community. In fact, one of the most developed BOKs is the PMBOK® guide, which provides a framework of best practices for project management [50]. The PMBOK® guide describes the processes, knowledge areas, and techniques that are necessary for successful project management. Also included is a glossary of commonly used terms in the project management field, as well as some key competencies and skills required for successful project management. The IMBOK is another well-developed BOK that describes information management principles, practices and skills [51]. This BOK defines some knowledge areas as well as key words and tools that can be used for information management. The SWEBOK, on the other hand, provides a description of the knowledge and skills needed to develop quality software systems [52]. It covers a broad range of software engineering topics, including software configuration management, software engineering management, and software engineering processes. It also provides

several tools that software developers can use to effectively develop their software. Finally, the EnVEBOK highlights the knowledge, skills, and competencies needed to practice environmental engineering practice [53]. It also describes a variety of tools and techniques that are important to environmental engineers.

By comparing the structure and information found in these BOKs, several common elements were identified, and can be used in the development of the proposed BOK for DfD/DfR in this study. The elements include the framework used to develop the BOK, the knowledge areas related to the subject matter, terminology related to the domain, essential competencies and skills needed in the field, as well as tools and techniques commonly used in the related domains. Table 1 illustrates these common elements in detail for each of the BOKs mentioned above.

2.5. Synthesis in the literature review

The literature review showed that the consumption of high-tech products is increasing, which means that more and more of these products are reaching their end-of-life and require proper management [24,54]. Among high-tech products, this paper focuses on electric and electronic equipment EEE that share several attributes, such as high degree of innovation, a commonly short lifespan, and the use of different materials including critical and rare materials [20–23]. The literature has also highlighted the importance of integrating eco-design practices from the early stages of high-tech product development in order to reduce their environmental impact [31]. Indeed, the growing volume of Waste from Electrical and Electronic Equipment (WEEE) is an environmental burden that needs to be addressed [24]. Since these products are generally not designed with end-of-life in mind, recycling is the most widely used technology to date for achieving a circular economy. The review also revealed that recycling could be improved when the disassembly of these products is considered during the design process [29]. In fact, DFD also needs to be considered to achieve a better circular economy [36,37]. There is also a lack of knowledge in the implementation of eco-design practices in high-tech companies that usually need skilled people to develop innovative products [40]. Therefore, DFD and DFR are of high importance in extending the lifecycle of EEE and minimizing their environmental footprint [36,37]. However, these practices are not widely adopted in companies, due to the lack of knowledge [38], and the development of a BOK for DFD and DFR of high-tech products is therefore needed to help fill this gap.

Various BOKs have already been developed in different fields [42–49], and our review allowed us to identify key elements in each of these BOKs. While no BOK directly related to eco-design or to high-tech

Table 1. Common structural elements between BOKs. Synthesis done based of the information from [43,50,52,53].

EnvE BOK	This BOK designed around an outcome-based structure, with a focus on the competencies that environmental engineers should possess	specific learning areas that are critical to achieving the outcomes, such as mathematics, conservation of energy and economics	definitions of many key terms and concepts, such as environmental engineering and sustainability	describes a wide range of competencies and skills, such as management and communication skills, business knowledge and leadership	includes several tools and techniques that are important for env. engineers, such as programming languages and software for graphics, GIS, and modeling.
PMBOK®	process-based framework that outlines the various project management processes	knowledge areas that are essential for successful project management, such as project integration management, project scope management, and project schedule management.	includes a glossary of terms and definitions, as well as standard set of abbreviations and acronyms commonly used in project management	three key competency areas for project managers: technical project management, leadership, and strategic and business management	techniques that can be used in project management to improve project performance such as planning techniques, estimating techniques and risk management techniques.
IMBOK	This BOK is based on Allen lee's model and built around six management segments	KA related to information management, such as information technology, information systems, business processes, business information and business strategy	defines key words such as 'entity', 'information', 'activity', 'process'	identifies several competencies and skills that are important in information management, such as technical skills, business skills, and analytical skills.	includes strategic analysis tools that can be used in information management such as SWOT analysis, five forces analysis, and value chain analysis.
SWEBOK	This BOK consists of ten knowledge areas, which cover the fundamental concepts and principles of software engineering	KA related to software engineering, such as software requirements, software design, software construction, software testing, and software maintenance.	Defines a variety of terms and concepts commonly used in the field of software engineering, such as requirements, design, testing, maintenance, and quality assurance	set of competencies related to the knowledge areas, such as the ability to create software design, the ability to develop software code and the ability to modify software systems	defines a set of tools and techniques that software engineers can use, such as software requirements tools, software design tools, and software construction tools.
common elements	Framework	Knowledge areas	Terminology	Competencies/skills	tools/techniques

products could be found, the comparison of identified elements allowed us to define common elements that could be used in any BOK. These key elements could be summarized in the framework, knowledge areas, terminology, competencies, and skills, as well as tools and techniques. It should be noted that these elements do not necessarily constitute an independent section in all BOKs. For instance, terminology is included in other sections of the IMBOK where definitions are given in the context of explaining the subject information. However, including an independent part of each element contributes to the clarity and understanding of our BOK. The following paragraph will discuss in more depth the framework used to develop the DFD and DFR BOK for high-tech products.

3. Method: the proposed BOK for DFD/DFR of high-tech products

As mentioned earlier, the comparison between several bodies of knowledge allowed us to identify common elements among them, which provides a basis for the development of our BOK for DFD and DFR of high-tech products. The BOK is illustrated in Figure 1.

3.1. Part 1: knowledge areas

The first key element of the proposed BOK is the knowledge areas related to DFD and DFR, which

includes 1) DFD and DFR guidelines that are essential to developing an eco-friendly product, providing standards and principles to make sure that the product is designed and manufactured with recycling and disassembly in mind, 2) second life applications of the product, a key element that helps designers become familiar with the potential reuse or repurposing of the product, so that they incorporate this knowledge into their design process and create a product with a reduced environmental impact, 3) recycling technologies that enable designers to develop products that can be easily recycled and therefore reducing the amount of waste going to landfills, and finally 4) knowledge of manufacturing processes that helps designers understand how the products are manufactured, and design products that can be easily disassembled and recycled using existing technologies with minimum environmental impacts.

3.2. Part 2: tools and techniques

The second key element of this BOK is the tools and techniques, as they help design high-tech products in ways that promote sustainability and reduce their environmental impact. Life Cycle Assessment LCA is the first technique to be mentioned, as it helps in assessing the environmental impact of the product throughout its life cycle, and guide designers toward a sustainable material selection, manufacturing

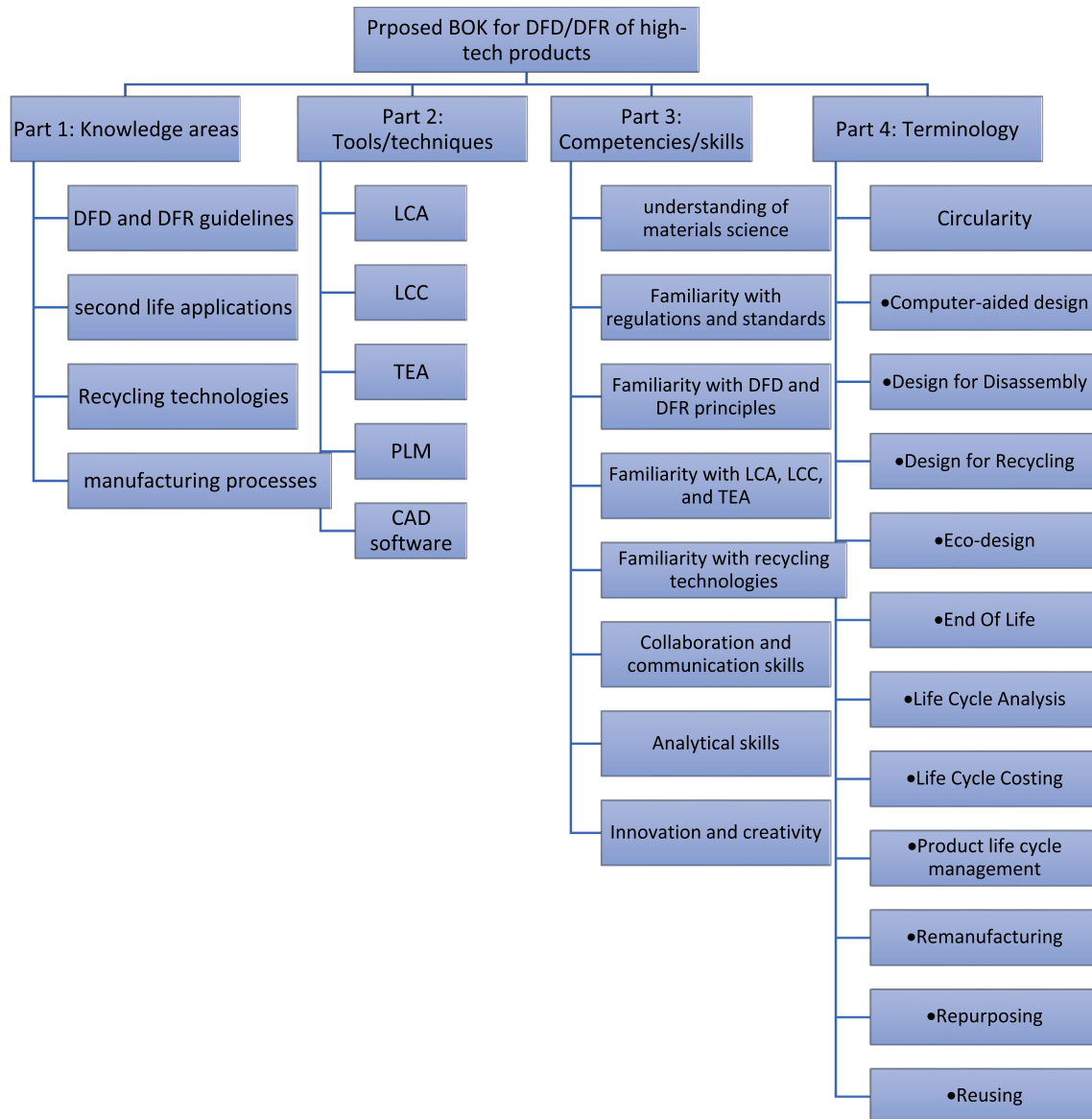


Figure 1. The proposed BOK for DFD and DFR of high-tech products.

processes, and end of life scenarios. In addition, Life Cycle Costing LCC is an important technique that helps in identifying the total cost of a product, from design and manufacturing to disposal, allowing designers to identify opportunities to reduce costs and improve the sustainability of the product. Along with LCA and LCC, Techno-Economic Analysis TEA is an important tool for DFD and DFR, since it allows an analysis of the economic impacts of a product from raw material extraction to disposal and can help identify the potential cost savings associated with DFD and DFR. Moreover, Product Lifecycle management PLM is a strong tool that helps designers control a product from its initial design to its final disposal and makes sure that the product is developed in alignment with DFR and DFD rules. Another important tool for DFR and DFD is Computer-Aided Design CAD software, which enables the design and visualization of product models in 2D and 3D and helps optimize them for improved disassembly and recycling. Moreover, the adoption of CAD

helps in accelerating the process of innovation, encourages automation of manufacturing facilities, and helps designers communicate with manufacturing staff, companies, and potential users for a better design for sustainability [55]. For instance, a case study carried out at an Indian manufacturing company showed that using CAD to optimize the components of a product minimized its environmental impact [56].

Table 2 summarizes and highlights the contribution and Key Performance Indicators (KPI) of the above-mentioned tool.

3.3. Part 3: competencies and skills

Competencies and Skills are the third key component of our BOK, as they are essential to DFD and DFR success. First, an understanding of materials science helps designers to decide which materials to use for improved disassembly and recycling. Second, the familiarity with regulations and standards related to

Table 2. Contributions and KPIs of tools and techniques, adapted to DFD and DFR from [57–61].

Tool	Contribution	KPI (Key Performance Indicators)
LCA	Compares relative environmental performance of different disassembly or recycling options	Energy consumption, Pollutants, resources etc. related to the design alternatives.
LCC	Determines cost-effectiveness of alternative disassembly or recycling option	Cost and benefit monetary flows related to the design alternatives.
TEA	Analyzes the technical and economic performance of different disassembly or recycling options	Assessment of investment alternatives and comparing them to existing processes.
PLM	Links production information to design, enabling comparison and optimization of different product DFD and DFR requirements	Quality KPIs, Time KPIs, Cost KPIs, Infrastructure KPIs and communication KPIs
CAD	Create/modify the product's geometry and structure	Design efficiency, flexibility, visualizing sustainability.

DFD and DFR ensures that products are developed in accordance with these regulations and standards about environmental sustainability and safety, such as restrictions on the usage of dangerous materials. Third, familiarity with DFD and DFR principles enables designers to adequately implement these concepts into the product design process, creating products that are optimized for both disassembly and recycling. An additional skill to have been familiarity with LCA, LCC, and TEA, as these tools provide a structured approach to assessing the ecological and economic impacts of the product over its entire life cycle. Moreover, as recycling technologies are constantly improving, the familiarity with the recycling technologies allows designers to be aware of the latest technologies and how to use them to their advantage when designing a product optimized for disassembly and recycling. In addition, DFD and DFR generally involve a variety of stakeholders with multiple expertise and perspectives, thus collaboration and communication skills are essential. These skills ensure effective communication between the parties to ensure that the product is designed for efficient recycling and disassembly. Analytical skills are important as well since DFD and DFR require analysis of the products design and materials to understand the impacts of design choices on the disassembly and recycling of high-tech products. Finally, designers should be innovative and creative, because they must have new ideas and strategies to improve the recycling and disassembly of the products they design.

3.4. Part 4: terminology

The fourth and final important element of this BOK is the terminology, which is the definition of commonly used keywords related to the DFD and DFR of hi-tech products. This terminology is intended to create a common language and understanding among professionals to facilitate communication and improve the efficiency of the DFD and DFR processes. It also decreases the effort and time needed to explain technical concepts. This terminology includes:

- **Circularity:** Achieving a circular economy where materials and products are used for the longest possible period.

- **Computer-aided design:** Designing products using software tools to improve the design process.
- **Design for Disassembly:** Designing a product in a way that makes it easier to disassemble.
- **Design for Recycling:** Designing a product while choosing materials that could be easily recovered at end-of-life.
- **Eco-design:** Designing products that are environment friendly, to reduce their environmental impact during their life cycle.
- **End Of Life:** The stage in a product's life cycle when it can no longer be used.
- **Life Cycle Assessment:** A method used to assess the environmental impact of a product throughout its life cycle.
- **Life Cycle Costing:** A method used to assess the cost of a product over its life cycle.
- **Product life cycle management:** A process that manages the entire product lifecycle, from initial design to end-of-life.
- **Remanufacturing:** Modifying used products to a like-new or better condition, to extend their life.
- **Repurposing:** the process of using a product or material for a different purpose than it was originally intended.
- **Reusing:** the process of using a product or material for the same purpose as it was originally intended

4. Application and results: the case of lithium-ion batteries

4.1. Lithium-ion battery and its applications

A LIB is a rechargeable battery that functions by the movement of lithium ions between the cathode and the anode, through an electrolyte [62]. These Batteries are gaining in popularity due to their high energy and power density, long-life cycle, as well as environmental friendliness, and they are mostly used in mobile devices and electric cars [63–66]. In this study, the focus is on LIBs used in electric vehicles and Figure 2 shows the key components of a LIB, where several electrodes (cathode/anode) alongside the electrolyte, separator and cell container form a battery cell, and different battery cells comprise a module. The

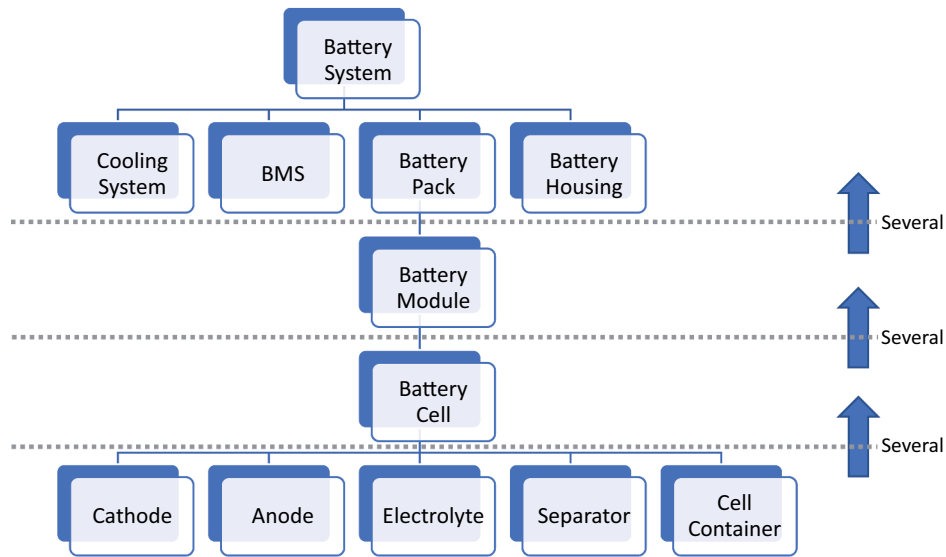


Figure 2. Main components of a lithium-ion battery.

modules, when combined, form the battery pack, which is a key component of the battery system, along with the cooling plates, cooling system, battery management system (BMS), and battery housing. The electrolyte plays an important role in ion transport [67]; while charging the battery, lithium ions move from the cathode to the anode, and vice versa during the discharge phase [68]. The role of separators is to prevent the anode and cathode from touching, and thus to avoid a short circuit of the battery. The operation of the EV battery pack is maintained by the BMS, in terms of charging time, safety parameters and life span. It is important to note that different cell formats exist, such as pouch, cylindrical, or prismatic, which are chosen depending on the application [69,70]. In addition, LIBs are named depending on their cathode materials, and Figure 3 illustrates the typical cell materials used with their corresponding LIB chemistry name names. A key observation is that the architecture of LIB cells, modules, and packs varies significantly from one manufacturer to another [71].

4.2. Lithium-ion batteries as an example of high-tech products

LIBs were chosen as an example of high-tech products because they share several characteristics of these products:

- The number of Li-ion batteries on the market is significant, their expected growth rate is high, and they are being rapidly integrated into the automotive industry. In fact, LIBs demand is expected to increase by approx. 33% annually to reach around 4,700 GWh by 2030 [72]. Moreover, Electric Vehicles manufacturers are showing great interest in LIBs, which started to steadily supersede nickel-metal hydride batteries generally used in the automotive industry [73,74]. In addition, Electric Vehicles are expected to account for 30% of all passenger cars worldwide in 2032 [75], with today's electric vehicles nearly entirely powered by LIBs [76,77]
- They contain toxic and strategic materials such as Cobalt, Lithium, Nickel, and Graphite. Cobalt, which is essential for the cathodes in some LIBs technologies such as LCO, is considered a strategic material because of the risk of supply interruption in the short and medium term [78]. Moreover, Nickel, critical material for the cathodes in NMC and NC, alongside graphite, mainly used in the anodes, as well Lithium and Cobalt, are four elements that are classified as critical for the US and EU ([79,80]. To be noted that batteries are generally classified as hazardous waste due to their toxic, reactive and combustible nature [81]
- A high level of innovation is needed to develop better LIBs; It is believed that advancements in LIB technologies require innovative chemistries for the electrolyte and electrode components [62]. Also, the manufacture of innovative inactive cell components is important to best meet the needs of various battery applications [82].
- End-of-life constraints are not usually taken into account while developing LIBs, as current LIB designs are optimized for cost, performance, and safety [83]

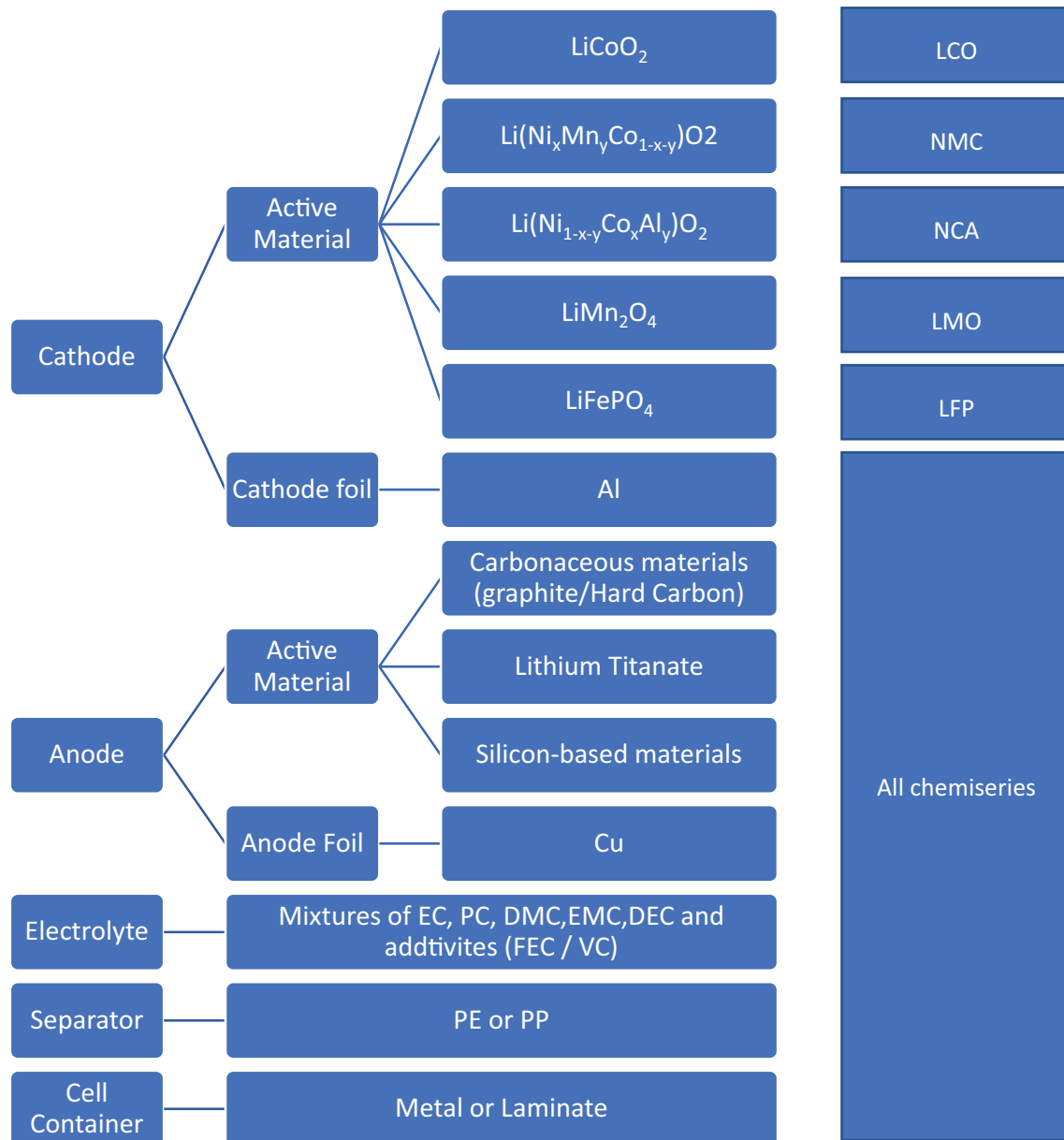


Figure 3. Typical LIB cell material, modified from [29].

4.3. Proposed BOK for DFD/DFR of LIBs

In this section, the practical application of the proposed BOK will be demonstrated by concentrating on LIBs. In doing so, its relevance and effectiveness in supporting sustainable battery development will be highlighted. This discussion explores various areas of knowledge within the industry, explore which tools are commonly used for the development of greener batteries, and focus on the key competencies and skills required for the effective design of disassembly and recycling processes. To do so, a detailed review was conducted, and the information needed was extracted from scientific articles as well as from the web.

4.3.1. Part 1: knowledge areas

4.3.1.1. DFD and DFR guidelines. These guidelines are imperative to facilitate designers in enhancing the design of LIBs in accordance with DFD and DFR

principles. The formulation of these guidelines has relied upon the DFD principles established by [84], which consists of five major “families”, namely 1) materials, 2) Fasteners, joints and connections, 3) Characteristics of components for disassembly, and 5) disassembly conditions. Each of these families includes different rules for improving the DFD. In addition, the research on DFR of LIBs showed that the rules related to DFR could be integrated into the same families above, since they consist of almost material-related rules. Table 3 provides an overview of the important DFD and DFR guidelines for LIBs and classifies them into the five big families for more precision and clarity. It is also important to add a new rule to the *product structure* family, which is to improve component design, to help cover all DFD and DFR rules for LIBs. Lastly, a color-coded scheme has been implemented to distinguish between DFD rules, DFR rules, and rules



Table 3. DFD and DFR guidelines about here.

DFD/DFR rules		Fasteners, joints and connections						Characteristics of components for disassembly		Disassembly conditions	
Product structure		Materials		general rule		LIB specific rule		general rule		LIB specific rule	
general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule
Create a modular design	- separable cooling system - modular construction - Module-less packs with cells that are easy to separate and open	- Minimise the use of different materials	- minimise the use of different materials within product and parts - minimizing the number and length of wires and cables - Rigid polymer exteriors are often well-suited to recyclability. Easy removal - Standardization of materials - Minimize non-reversible adhesives - minimize hard to recycle materials - Characterising the binder distribution and adhesion - purity of cathode material - Develop technologies to recover lithium of pyrometallurgical process - Develop technologies to enable recovery of electrolyte and graphite in hydrometallurgical process - Effectively separate cathode material from other materials	- Minimise the number of joints and cnx	- standardisation in fixing type	- Good accessibility	- The first design issue that needs to be addressed is how to open the pack, module and cell easily	- Design for automated disassembly	- adjust design towards simple dismantling		
minimise the component count	- overall part number	use recyclable materials		make joints visible and accessible; eliminate hidden joints, and mark non obvious joints	low weight			eliminate the need for specialised disassembly procedures	- number of operations required in order to reduce the battery to the required scale		
optimise component standardisation	-Standardisation of pack opening - Standardization of product size and design	- eliminate toxic or hazardous materials	- Electrode materials: Co less/free, organic materials, single-crystal to boost stability of Li-rich and Ni-rich - Replacing flammable carbonates-based electrolytes with recyclable aqueous and solid-state electrolyte - use water soluble adhesives - Avoidance of toxic materials (cadmium, arsenic, mercury, halogens, etc.)	eliminate hidden joints	robust, minimise fragile parts			DFD with simple and std tools	- standardisation in fixing type would simplify disassembly as it would only require one tool		

(Continued)

Table 3. (Continued).

Product structure		Materials		Fasteners, joints and connections		Characteristics of components for disassembly		Disassembly conditions	
general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule	general rule	LIB specific rule
minimise product variants	Large number of designs and manual disassembly	use durable and not corrode	material for labels	use joints that are easy to disassemble, such as nuts and bolts	- glued and welded joints are used - use two-way snap fits or break points on snap fit - locate snap fits so that they can be opened using standard tools - use joining elements those are easily separable or easy to destroy	preferably unpainted			
improve Component design	- changing the geometry of the electrode connector tabs - use of solid-state electrolytes - Anode-free battery, free-standing electrode without a binder, and water-soluble binders - Quantitative labelling => recommended disassembly and recycling - Ensure chemical compatibility between the exterior battery materials and the recycling process - use of molded-in colors instead of paints - design fastening points easy to access - provide identification for separation points - Design parts so that they there is no need to separate them			use fasteners rather than adhesives	- Avoid potting or adhesive compounds to hold cells in place - use fasteners of compatible material with the parts being joined				

applicable to both domains, where DFD rules are presented in black text, DFR rules in green text, and common rules in red text.

4.3.1.2. Second life applications. Given that used batteries for electric vehicles still contain 80% of the primary energy [85], second life applications should be considered to optimize their DFD and DFR, extend their lifecycle and thus minimize their environmental impact. Based on the State of Health (SOH) of used EV LIBs, HSSMI proposed an end-of-life strategic decision tree [86], which shows the second life applications of a used LIB : Reuse, Repair, Remanufacture, Repurpose and Recycling. This decision and tree is represented in Figure 4 alongside the waste hierarchy of LIBs [87]. While reuse means that the battery can be reused for the same purpose for which it was designed, typically after repairing it, repurposing means that the used battery could be used in a different application than the one for which it was originally designed, such as residential, grid, or renewable power plants. Remanufacturing, on the other hand, means that the used batteries can be restored to at least their original efficiency, or even better. It should be noted that remanufacturing, repurposing, and recycling are the most used in the industry, with recycling being the ultimate option [88]. To get the most out of LIBs, the optimal scenario would include reuse, remanufacturing, or repurposing first, followed by recycling [86,89,90]. While reuse, remanufacturing, and repurposing extend the life of LIBs, recycling is the destination to recover their basic material, such as lithium,

cobalt, or copper. The first option only postpones the recycling timeframe.

4.3.1.3. Recycling technologies. Recycling of EV LIB packs involves a combination of chemical, mechanical, and thermal processes to recover the various valuable target materials available in batteries such as cobalt and nickel [91]. Three recycling technologies exist: Pyrometallurgy, Hydrometallurgy, and direct recycling, the latter being the ideal option but still at the laboratory and pilot scale. The most important component of the pack in terms of materials is the LIB cell, and the recovery of cathode materials is the main objective of the recycling process [83,92,93]. Advantages of hydrometallurgy compared to pyrometallurgy include lower energy consumption, higher extraction rates, reduced capital investment, and fewer hazardous gas emissions. However, its main drawback is its low adaptability to various material compositions of different LIBs types [94]. It should be noted that each recycling technology requires the battery to be disassembled to a certain level, the added value of material recovery increasing with a higher level of disassembly (disassembly to cell level for example), but technological readiness of the process simultaneously decreasing [95]. At present, waste batteries suffer from high disposal costs, low recycling efficiency, and limited recycling profits. As mentioned, disassembly plays a major role in recycling efficiency, especially when opening a LIB pack is a complex task. Indeed, the pack can be deformed following a car accident for example, the joints can be rusted, and above all, the packs come in

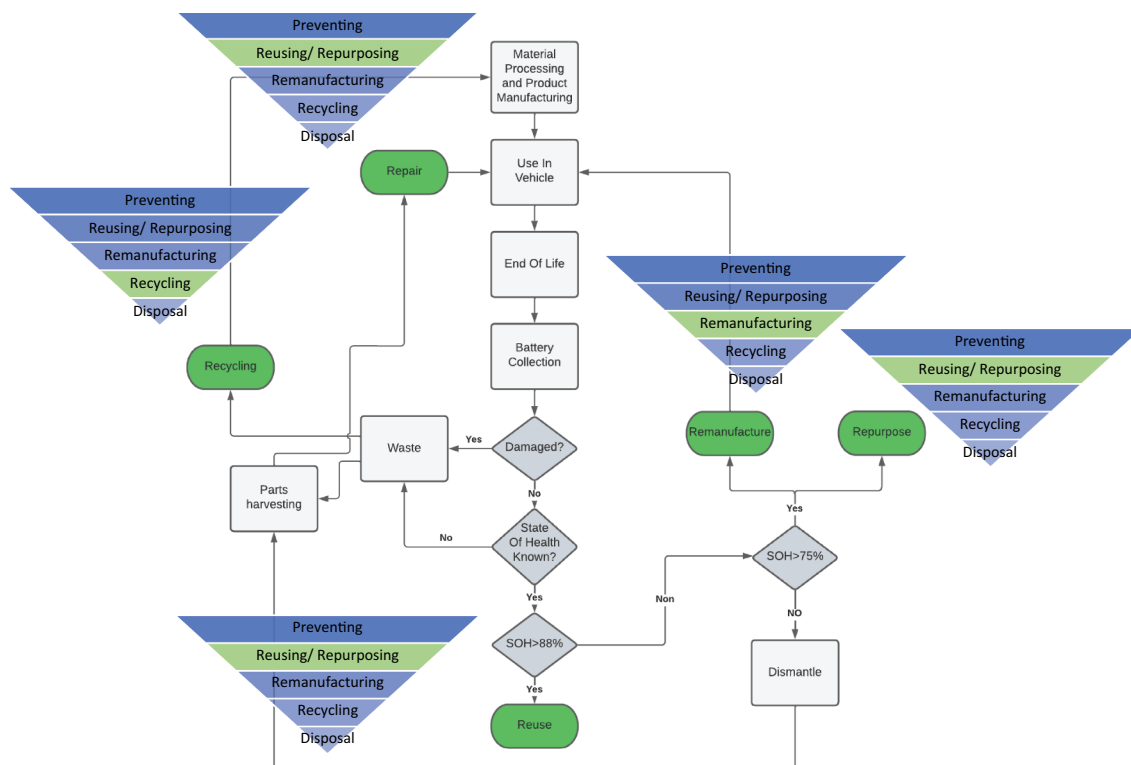


Figure 4. LIB second life application, modified from [86,87].

Table 4. Example of regulations and standards of LIBs.

Application of regulations and standards	Regulation	Origin	description
Environmental considerations	Resource Conservation and Recovery Act (RCRA) Laws and Regulations.	U.S.A.	Under the Resource Conservation and Recovery Act (RCRA), the U.S. Environmental Protection Agency (EPA) has regulations governing the management and disposal of hazardous waste, including certain types of LIBs [111].
Compliance with legal requirements	Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL concerning batteries and waste batteries, repealing Directive 2006/66/EC, and amending Regulation (EU) No 2019/1020.	European Union	This proposed regulation highlights the need for sustainable design practices, including a focus on disassembly and recycling of LIBs [112].
Worker's safety	1910.1200 - Hazard Communication standards	U.S.A.	Employers should communicate the hazards associated with chemicals, including LIBs, through labels for examples, which can be applied during the design phase [113].

different shapes, sizes and material, and the accessories used are not standardized [83]. Also, the complexity of pack disassembly plays an important role in the recycling cost, especially that the manual disassembly of these batteries entails high costs and long lead times [91,96,97]. The disassembly in the future research should focus on advanced technologies such as artificial intelligence (AI), machine learning (ML), and Internet of things (IoT) that can enable waste traceability and real time monitoring of the battery status. Moreover, automated disassembly – which is still in early stages – can improve the efficiency and profitability of recycling [98].

4.3.1.4. Manufacturing processes. Manufacturing processes often involve automation and assembly line operations. Knowing that lithium battery manufacturing requires automation due to the high precision required [99], designers can use this knowledge to match existing automation techniques in recycling plants. By considering the limitations of automated disassembly processes, LIBs can be designed for efficient and cost-effective recycling. In addition, several methods exist for different manufacturing processes of LIBs, such as laser welding or wire bonding for the welding process [100]. However, the welding process contradicts the guidelines mentioned in the paragraph 3.1 of the BOK, and designers can think of innovative manufacturing processes that are in accordance with the guidelines, to improve the disassembly and recyclability of LIBs. Moreover, the knowledge of manufacturing processes helps in designing LIBs with modular and easily separable components. By considering the assembly methods and techniques used during manufacturing, designers can ensure that the battery components can be efficiently disassembled for recycling without excessive effort or damage. Finally, Manufacturing processes provide insights into the materials and their compatibility within the battery assembly. This knowledge enables designers to select materials that are suitable for disassembly and recycling processes.

Some materials can be difficult to separate during recycling, such as those with strong adhesive properties or complex composite structures. By understanding the manufacturing processes, designers can avoid these materials or find more easily recyclable alternatives.

4.3.2. Part 2: tools and techniques

LCA has been widely used in the past years to study the environmental impact of the LIBs throughout their lifecycle. It has been found that the life cycle impacts of these batteries depend highly on the chemistry used; the materials used in batteries and the related production methods vary according to the particular chemistries and play a key role in their life cycle impact during the manufacturing and disposal stages [101]. For instance, in NMC and LFP batteries, the active cathode material and energy use for cell manufacturing have the most environmental impact [102,103]. Another study found that in LIBs, it's preferred to use water as a solvent instead of NMP, and that the internal battery efficiency is as important as the battery weight in terms of environmental impact, if not better [104]. It should be also noted that several LCA studies have shown that batteries with longer life expectancy and higher energy density have lower negative environmental impacts [105,106]. Finally, a critical review on LCAs of LIBs found that the most analyzed chemistries are LFP, LMO and NMC, and that there's a lack of primary data used in these studies, especially for energy consumption during the production phase [107].

While LCA is an important tool for understanding the environmental impact of LIBs, LCC and TEA are also used to address the economic aspect of DFD and DFR in LIBs. In fact, LCC can provide valuable information on the economic feasibility of various design options for disassembly and recycling of LIBs, from initial costs to operating and maintenance costs to disassembly and recycling costs. In addition, TEA can help assessing the economic feasibility and competitiveness of different recycling technologies (Pyrometallurgy, Hydrometallurgy, and direct recycling) to select the

most cost-effective recycling method, estimate the capital and operations costs and potential revenues, and optimize the recycling processes.

PLM systems provide a robust infrastructure for managing product data, facilitating collaboration, ensuring regulatory compliance, and supporting decision-making throughout the entire lifecycle of high-tech products. There are different challenges in DFD and DFR of high-tech products since a variety of models and tools should be used at the design stage and different databases including technical, environmental, and economic data should be integrated. Hence, visualization and data analytics aid the eco-design team in having a systematic and holistic approach to DFD and DFR [108]. The decision tools should be developed to integrate the massive data and apply advanced simulation, and optimization provides multi-variant analysis. Figure 5 shows the type of data should be analyzed for DfD. By integrating DFD and DFR considerations into the PLM platform, designers can effectively incorporate circular economy principles and promote sustainable practices in LIB design and manufacturing processes. PLM data can aid designers to select materials that are nontoxic, high-quality, and recyclable and optimize the product geometry and structure for easy disassembly and reuse, and to track the environmental impacts and recovery values of the product and its parts. PLM data can also facilitate information sharing and collaboration

among different stakeholders involved in the product design, manufacturing, operation, and end-of-life stages.

Computer-Aided Design (CAD) is also beneficial for the DFD and DFR of LIBs. It can help visualize the battery design, which allows the design to be optimized to facilitate disassembly and recycling processes (type of material, joint types etc.). Moreover, it helps designers create disassembly sequences for LIBs and identify potential challenges or inefficiencies in the disassembly and recycling processes. Furthermore, these tools improve collaboration and communication between different stakeholders, ensuring that disassembly and recycling are taken into consideration during the design phase. Finally, the integration of circularity indicators into CAD software allows designers to assess and optimize the sustainability and circularity aspects of the battery design.

4.3.3. Part 3: competencies and skills

4.3.3.1. Understanding of materials science. As mentioned earlier, LIBs come in a variety of shapes, sizes, and chemistries. It is critical that designers understand materials science to be able use materials that can improve the recycling and disassembly of these batteries. The focus here is on battery cell materials, as these are the most valuable in LIBs recycling. The guidelines presented in the Part 1: Knowledge areas cover a comprehensive overview on material related

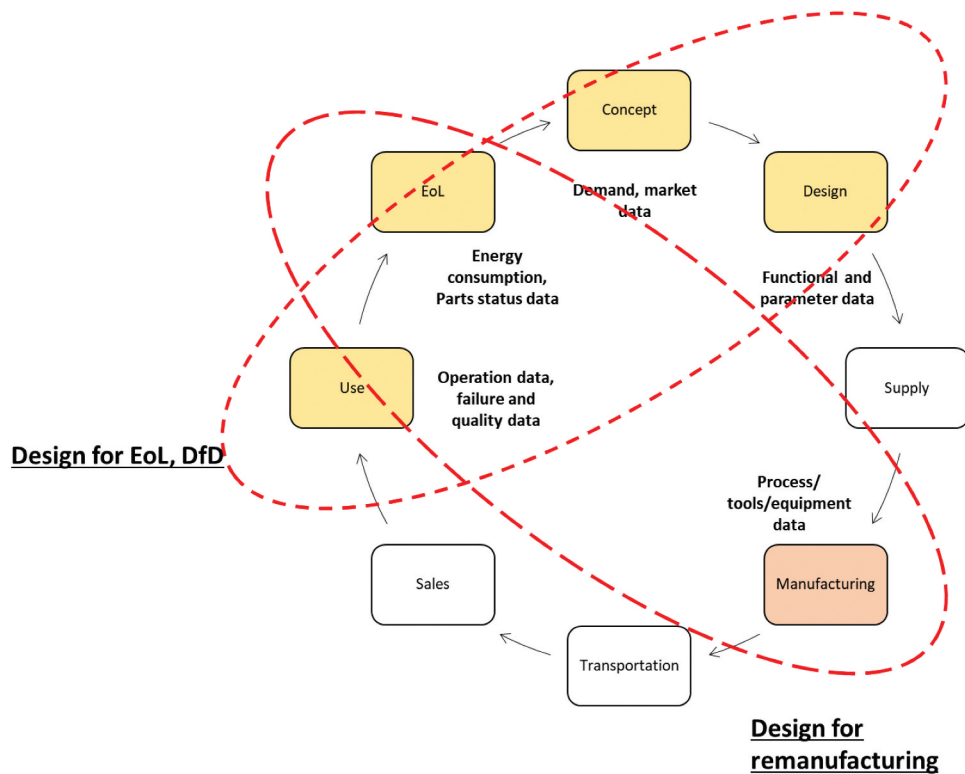


figure 5. Type of data needed to integrate DFD in PLM.

actions to be taken to improve the recyclability and disassembly of LIBs, such as standardization of materials and minimizing the hard to recycle materials and non-reversible adhesives. Understanding materials science helps designers in the material selection process so that they use materials that are easier to disassemble and recycle and optimize their properties to ensure a high-performance battery. Understanding materials science also allows designers to select fasteners, adhesives or other bonding techniques that make it easier to disassemble the battery without damaging it. For example, a major challenge for disassembly is the collection of active materials from the electrodes by dissolving the binder, typically polyvinylidene fluoride (PVDF). However, this binder can only be dissolved in a few solvents such as N-methyl-2-pyrrolidone (NMP), which is expensive and risky. Therefore, new binders are proposed, such as styrene-butadiene rubber (SBR), carboxymethylcellulose (CMC) and polyethylene (PE) [109]. In addition, debondable adhesives could be used instead of the typical structural adhesives for faster disassembly and recycling. In fact, these adhesives can be debonded when applying an external stimulus, and in the case of LIBs, can be effectively used at a cellular level. However, their use in pack and module level is not applicable due to the materials used and service conditions [110]. A final note is that there's a big gap between materials research and materials scale-up for large scale LIBs manufacturing, and an effective integration of materials science, engineering, chemistry, and analytical tool is essential for developing innovative battery materials [82].

4.3.3.2. Familiarity with regulations and standards. The familiarity with regulations and standards is critical to the design of LIBs, particularly in the recycling and disassembly context, for several reasons:

- Compliance with legal requirements: Companies must comply with the regulations established by policymakers regarding the handling, disassembly, and recycling of LIBs.
- Environmental considerations: Because LIBs include hazardous materials such as lithium, cobalt, and nickel, which pose an environmental risk if disposed of improperly, familiarity with regulations help in designing batteries that are easier to disassemble and recycle, minimizing their environmental impact.
- Workers Safety: Since LIBs can present safety risks during disassembly or recycling, due to flammable electrolytes for example, following safety standards that could be implemented during design can protect workers from potential accidents and hazards.

Table 4 illustrates examples of regulations and standards related to LIBs that fit in the above-mentioned applications. It's important to note that specific regulations and guidelines related to the DFD and DFR of LIBs can vary by country, so it's better to consult the regulations specific to the country in question.

4.3.3.3. Familiarity with DFD and DFR principles.

Understanding the principles of Design for Disassembly (DFD) and Design for Recycling (DFR) forms the fundamental basis for effectively managing sustainable practices concerning high-tech products. DFD entails the comprehension and incorporation of strategies that facilitate the organized dismantling of products at the end of their lifecycle, thereby promoting the efficient retrieval of materials. Conversely, DFR encompasses the strategic adoption and implementation of design characteristics that optimize the potential for product recyclability, ensuring the recovery of valuable resources through recycling procedures.

4.3.3.4. Familiarity with LCA, LCC, and TEA. A strong familiarity with life cycle assessment (LCA), life cycle costing (LCC), and technoeconomic analysis (TEA) is of paramount importance. Such knowledge empowers designers to assess the viability of DFD and DFR methodologies from environmental, economic, and technical standpoints throughout the lifecycle of the product.

4.3.3.5. Familiarity with recycling technologies. A comprehensive awareness of both established and emerging recycling techniques is essential for designers. These techniques encompass procedures like pyrometallurgy and hydrometallurgy, as well as cutting-edge technologies such as AI and IoT for real-time waste tracking and process monitoring. Possessing these competencies enables designers to make well-informed decisions regarding materials, product design, and end-of-life product management.

4.3.3.6. Collaboration and communication skills.

Successful implementation of DFD and DFR strategies for LIBs relies on effective collaboration and communication skills. These principles necessitate the collaboration of diverse experts, including battery manufacturers and recycling facilities, to create standardized designs that facilitate easy disassembly and recycling. Collaborative endeavors ensure a holistic perspective, ultimately optimizing disassembly and recycling processes. Effective communication is also vital for disseminating technical information and best practices among these varied specialists.

4.3.3.7. Analytical skills. During the phase of designing, the role of analytical skills is of utmost importance. Designers need to differentiate suitable

materials and elements that can be easily taken apart and recycled. It is also crucial to assess the effectiveness and economic viability of various methods for disassembly and recycling. Proficiency in analysis is essential for comprehending the environmental and financial consequences of DFD and DFR approaches. Designers must meticulously examine how diverse design options influence the entire lifespan of the product, enabling them to arrive at well-founded choices that align ecological deliberations with economic aspects.

4.3.3.8. Innovation and creativity. Thinking creatively and innovatively is important for finding new ways to make battery components easier to separate and identifying materials that can be recycled effectively. This innovative mind-set has the potential to give rise to the development of groundbreaking materials, components, and techniques for assembly, which in turn facilitate smooth disassembly and improvement of recyclability. Embracing inventive thinking and creativity empowers advanced technology enterprises to consistently refine their DFD and DFR strategies, thus contributing to the evolution of more environmentally friendly product designs.

4.3.4. Part 4: terminology

This section provides some additional terminology related to LIBs in the context of DFD and DFR:

- **Battery cell:** Basic building block of a LIB. Consists of anode, cathode, electrolyte, separator, and cell container.
- **Battery Module:** A battery module is composed of multiple battery cells arranged in a specific configuration.
- **Battery Pack:** A battery pack is a collection of individual battery modules assembled for a higher voltage and energy storage capacity.
- **Anode:** The negative electrode in a battery cell, where lithium ions are released during charging process and stored during discharging process.
- **Cathode:** The positive electrode in a battery cell, where lithium ions are released during charging process and stored during discharging process.
- **Electrolyte:** The electrolyte is the material that allows the movement of lithium ions between the anode and cathode during the charge and discharge cycles.
- **Battery Management System (BMS):** The Battery Management System is an electronic system that monitors and controls various parameters of a LIB, such as voltage and temperature,

- **State of Health (SOH):** State of Health refers to the current condition or performance level of a lithium-ion battery.

5. Discussion

In this case study, LIBs were chosen as an example of high-tech products because they share several characteristics with such products, such as their high number on the market and high growth rate, the significant level of innovation required, and the little attention given to their end-of-life management.

To justify the applicability of the BOK, the knowledge areas related to the LIBs were first examined; An original table of DFD and DFR guidelines was developed, consisting of important rules related to five major families: product structure, Materials selection, Fasteners, characteristics of component disassembly, and disassembly conditions. By incorporating these guidelines into the design process, designers can improve the recyclability and ease of disassembly of LIBs. The color-coded scheme used in the guidelines helps distinguish between DFD rules, DFR rules, and rules applicable to both areas, promoting clarity and understanding.

Then, second life applications of these batteries were presented based on their state of health. Reuse, remanufacturing, and repurposing were shown to significantly extend the life of LIBs, while recycling was the ultimate option to recover valuable materials. It is important to prioritize reuse, remanufacturing, or repurposing before recycling, as these options maximize the value and lifespan of LIBs. It is also essential to consider second life applications for used LIBs to optimize DFD and DFR, extend their lifecycle, and minimize their environmental impact.

Afterwards, it was found that three recycling technologies exist for LIBs: pyrometallurgy, hydrometallurgy, and direct recycling, and that they are used to recover valuable materials, especially those found in batteries cells, such as Cobalt and Nickel. It has also been shown that disassembly plays a crucial role in recycling efficiency, and that the complexity of pack disassembly can increase recycling costs. Future research should explore advanced technologies like AI, ML, and IoT to improve waste traceability and real-time monitoring so that proper actions can be taken during the design phase. Automated disassembly and standardization of battery pack shapes, sizes, and materials can enhance recycling efficiency and profitability.

The knowledge of manufacturing processes, on the other hand, helps designers improve the disassembly and recyclability of LIBs by considering the limitations

of automated disassembly processes and using innovative manufacturing techniques. Manufacturing processes also inform material selection and compatibility within the battery assembly.

Part 3.2 of this BOK, tools, and techniques, showed that the combination of LCA, LCC, TEA, PLM systems, and CAD tools provides a comprehensive approach to address the environmental and economic aspects of LIB design and recycling. These tools and techniques enable the selection of sustainable materials, optimization of products' DFD and DFR, and assessment of the environmental and economic impacts of LIBs throughout their lifecycle.

Part 3.3, Competencies and skills, argued that the understanding of materials science is crucial for designers involved in the recycling and disassembly of lithium-ion batteries (LIBs). By selecting materials that are easier to disassemble and recycle and optimizing their properties, designers can enhance the recyclability and performance of LIBs. Additionally, familiarity with regulations and standards is vital to ensure compliance with legal requirements, minimize environmental impact, and ensure worker safety during disassembly and recycling processes. Effective collaboration and communication skills, analytical abilities, and innovation are also essential for standardized designs, efficient methods, and the exploration of novel techniques that facilitate the disassembly and recycling of LIBs.

The fourth and final part described a terminology to be used alongside the general terminology developed in [section 3.4](#), which includes key words specific to DFD and DFR of LIBs, such as battery cell, battery module, battery pack and Battery Management System.

The study's theoretical implications go beyond immediate design practices. Integrating the principles of DFD and DFR into high-tech product design not only addresses urgent environmental concerns but also fundamentally reshapes the landscape of innovation. By embracing these principles, the study triggers a shift in how technology is conceived and developed, sparking a profound transition toward a regenerative and sustainable circular economy. This integration facilitates a harmonious synergy between technological advancement and environmental stewardship, exemplifying a new era of conscientious innovation. Furthermore, the article contributes to the existing literature on the eco-design of high-tech products by proposing a BOK framework for DFD and DFR. The BOK framework can help researchers and practitioners understand, apply, and evaluate DFD and DFR strategies in a systematic way. The article also demonstrates the application of the BOK framework to LIBs, showing how the BOK framework can be adapted and customized to different product types and contexts.

From a practical perspective, the study's findings offer actionable pathways for sustainable design. By implementing the provided DFD and DFR guidelines, manufacturers can directly enhance recyclability and disassembly efficiency of products like Lithium-Ion Batteries (LIBs). Furthermore, the study's insights into second life applications, recycling technology enhancement, and skill development equip designers with tangible strategies for reducing environmental impact while maximizing economic feasibility. The article provides practical advice and recommendations for manufacturers, designers, and engineers of high-tech products to implement DFD and DFR strategies from the earliest stages of product development. It also shows how the BOK framework can help assess the environmental impact and end-of-life performance of high-tech products.

Overall, this case study provided a structured framework, knowledge base, and practical guidance for designers to incorporate DFD and DFR into LIBs, integrate circular economy principles, and promote sustainable practices, ultimately contributing to a more sustainable and efficient management of LIBs throughout their lifecycle. It should be noted that each of the four parts of this BOK could be more detailed, by including different stakeholders such as OEMs, recyclers, and remanufacturers. However, for the purposes of this work, the level of detail presented is sufficient.

6. Conclusion

In summary, the literature review highlighted the importance of DFD and DFR for high-tech products. Therefore, incorporating DFD and DFR principles into these products enables efficient disassembly and recycling of e-waste, aiming to recover valuable materials and reduce environmental impact. Taking into consideration the importance of eco-design knowledge, the BOK for high-tech products DFD and DFR was developed. This BOK helps with sustainable design practices and enables the development of environmentally friendly high-tech products. The application of the proposed BOK to LIBs demonstrated its applicability and effectiveness in supporting sustainable battery development.

This article presents a limited scope regarding eco-design since it only covers DFD and DFR, whereas eco-design covers many more topics. In addition, it focuses on LIBs as a case study, which may limit the generalizability of its application to other industries. Also, the article is mainly based on secondary information, without including the various industry stakeholders, which may limit the variety of points of views discussed.

Extending this BOK could include an overview of all eco-design practices, not just DFD and DFR. Further research and development are also needed, involving various stakeholders in its development to gain varied

perspectives, promote collaboration among key actors in the value chain, and get practical insights and real-world validation. Considering additional case studies will also help to understand the application of DFD and DFR in different contexts, strengthening the validity of the proposed BOK, and addressing industry-specific challenges related to high-tech products. Finally, examining economic feasibility in more detail is essential for the practical implementation of DFD and DFR, and future research on this topic enables the article to provide more details on the financial implication of implementing DFD and DFR as well as proposing innovative business models that promote the principles of circular economy.

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Acronyms

BOK	Body of Knowledge
CAD	Computer-Aided design
DFD	Design for Disassembly
DFR	Design for Recycling
EEE	Electrical and Electronic Equipment
EoL	End of Life
EV	Electric Vehicle
LCA	Life Cycle Assessment

LCC	Life Cycle Cost
LIB	Lithium-Ion Battery
PLM	Product Lifecycle Management
TEA	Techno-Economic Assessment
WEEE	Waste Electrical and Electronic Equipment

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